

by small-scale physical changes into electricity. It has three typical approaches/types: piezoelectric, triboelectric and pyroelectric. **Both piezoelectric and triboelectric nanogenerators can convert mechanical energy into electricity, and are used in medical devices.**

A piezoelectric nanogenerator converts external kinetic energy into electrical energy via action by a nanostructured piezoelectric material such as ZnO, GaN, BaTiO₃ and the like.

A triboelectric nanogenerator (TENG) is an energy-harvesting device that converts external mechanical energy into electricity by a conjunction of triboelectric effect and electrostatic induction. As for this power generation unit, in the inner circuit, a potential is created by triboelectric

effect due to charge transfer between two thin organic/inorganic films that exhibit opposite tribo-polarity. In the outer circuit, electrons are driven to flow between two electrodes attached to the back of the films to balance the potential.

Triboelectric effect is a type of contact electrification on which certain materials become electrically charged after these are separated from a different material with which they were in contact. Rubbing a plastic pen with a polyester cloth can build up triboelectricity. Most everyday static electricity is triboelectric. Polarity and strength of charges produced differ according to the material, surface roughness, temperature, etc. Polyurethane foam is the most positively-charged, steel has no charge and ebonite is the most negatively-charged.

After coming into contact with other material, a chemical bond is formed between parts of the two surfaces, called adhesion. Charges move from one material to the other to neutralise their electrochemical potential. This is what creates the net charge imbalance between objects. When separated, some of the bonded atoms keep extra electrons, while some give them away.

e-Bandage

There are therapeutic effects of electrical stimulation on wound healing. It imitates the natural wound healing mechanism of the endogenous electric field to facilitate skin growth. An electric field is essential for directing many cellular processes that lead to orderly healing naturally.

Electrical stimulation decreases oedema around the electrode, stimulates growth of tissues, increases blood flow, causes fibroblasts to proliferate, induces epidermal cell migration and their growth, attracts neutrophils and macrophages, and so on.

The exceptional capability of the nanogenerator to generate periodic biphasic electric pulses by locally converting feeble mechanical displacements, such as body or muscle motion, is sustainable and biologically responsive. Besides, a nanogenerator can be made flexible with reasonably high output power density and energy conversion efficiency. Its multiple energy conversion mechanism, including both piezoelectric and triboelectric effects, allows a wide selection of materials and multiple design principles. Moreover, it is a very stable device.

The self-activated electrotherapy bandage device consists of two parts: biomechanical energy conversion part (that is, nanogenerator) and dressing electrodes made of gold (Au). As shown in Fig. 4, the nanogenerator is made by overlapping the copper (Cu)/PTFE electro-negative material layer with another copper (electro-positive) layer on different sides of the PET substrate.

The PTFE layer can slide back and forth relative to the copper layer, and when worn by the patient generates sufficient voltage for wound healing due to movement of the body. Due to good flexibility of PTFE, the multilayer device matches the skin surface. The device is non-toxic. Overlapping area change drives charge flow towards or away from the two dressing electrodes and, thus, induces electric potential between them, which is maintained until the skin regeneration process is complete.

The two dressing electrodes are placed, as shown in Fig. 5, on each

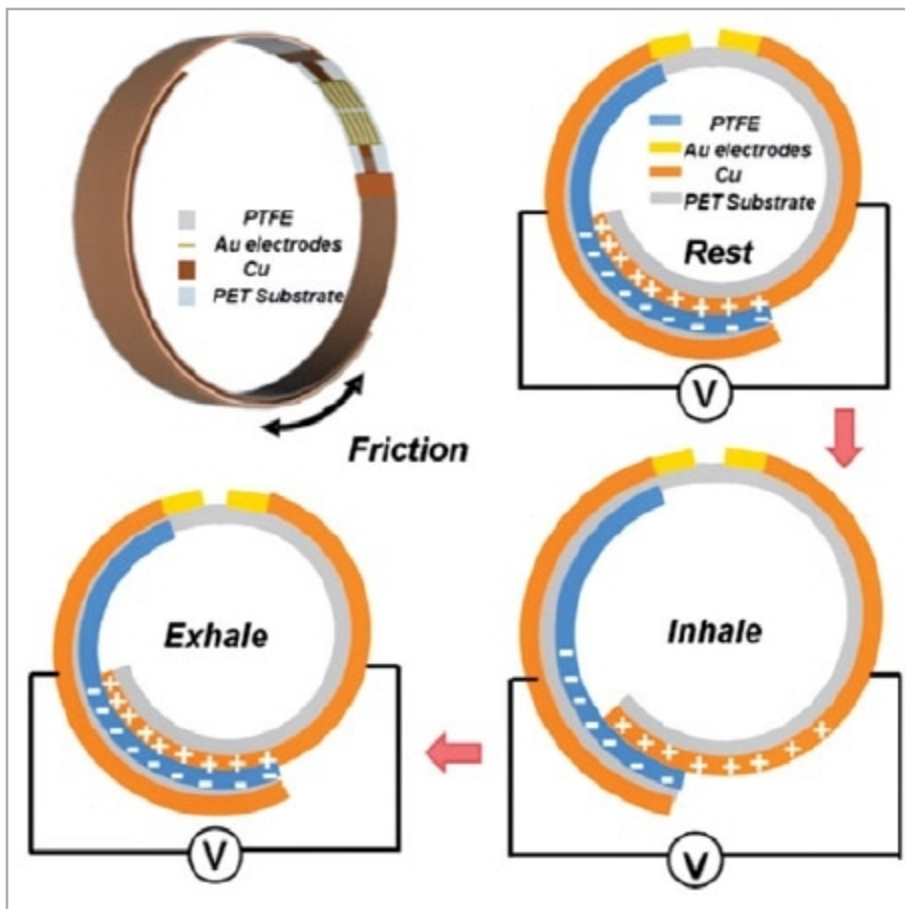


Fig. 4: e-Bandage

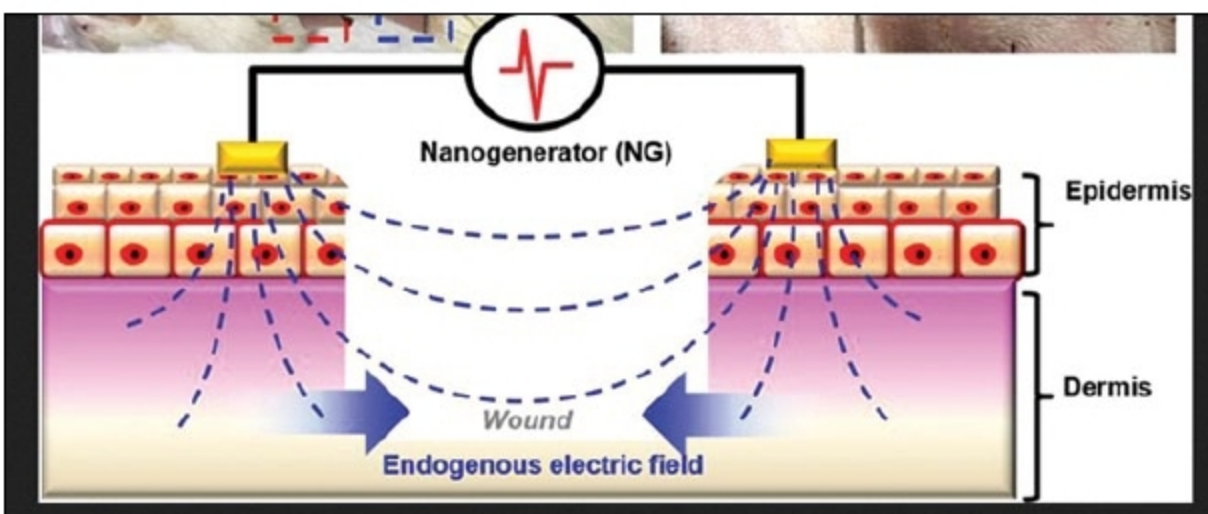


Fig. 5: Electric field in a nanogenerator